Navigation in unknown outdoor off-road envirronement using Interval Analysis

Celian DALIGAULT[12], Eric LUCET[2], Sebastien LAHAYE[1], Sebastien LAGRANGE[1], Remy GUYONNEAU[1]

[1]LARIS, Universite Angers — 62 avenue Notre-Dame-du-Lac 49000 Angers France
[2]nano-innov, CEA — 2 Boulevard Thomas Gobert 91120 PALAISEAU France

Context

Outdoor unknown off-road navigations stakes in mobile robotics application are multiples :

- Firstly, the unkown nature of the envirooonmeent necessitate to preform a full simulatuaous localisation and mapping (SLAM) whitout or with the least possible a priori hypothesis which is a well known and classical problem.
- The off-road specifity introduces many additional important information neccessary to know such as ground geometry, its nature (e.g. slipping ground) but also the presence and nature of dynamic objects to adjust the cartography models and potentially to avoid it when path generation is produced.
- In addition, when a robot evolve outdoor, it face evolutionarry envirronement condition like a change in the luminosity and wheater conditions. This change in the envirronement does not always allow to use continusously the same sensors. A choice in the sensors but also multisensor fusion is consequently essential.
- Moreover, the noisy nature of aquistion sensors must be a the subject of careful consideration to bring robust information from which path generation can be based on. The context of the described envirronement can easely correspond to applications such as construction or farming sites when vehicules evolve jointly with human operators, application when the security is the top priority. This last constraint justify how the use of the interval analysis should be considered.

A robust special sensor : The micro-wave RADAR

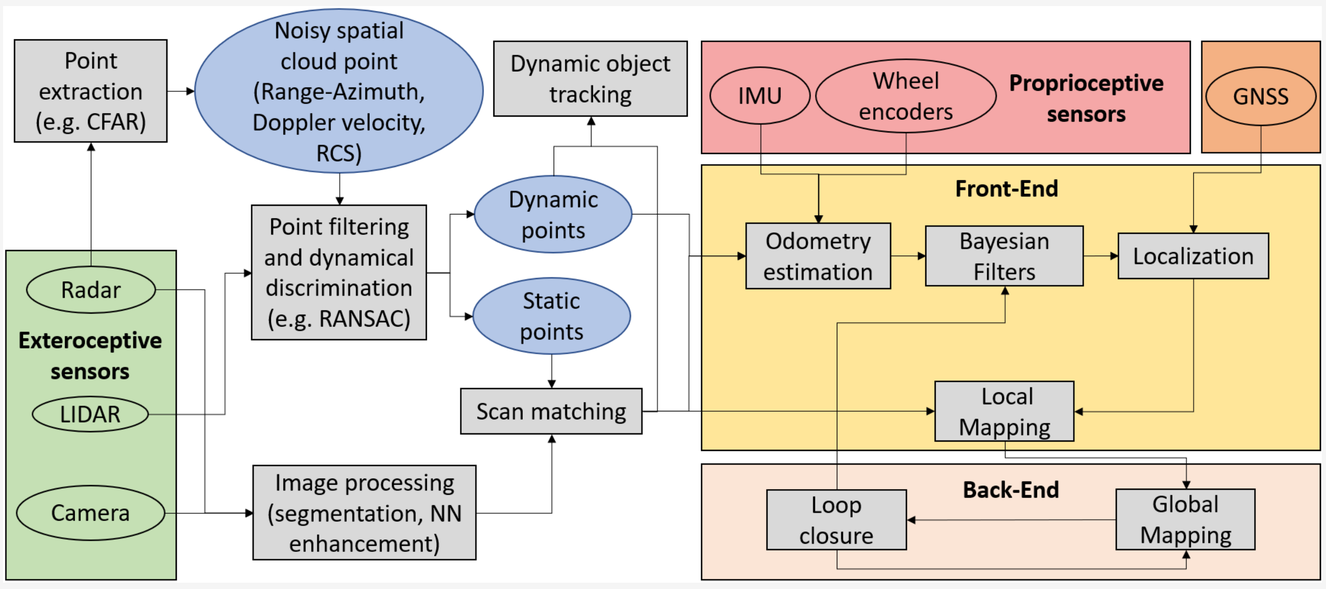

Figure 1. Overview of the radar integration in SLAM.

Similarly to the LiDAR, the RADAR is an electromagnetic based active sensor, which return the distance to an obstacle with the time of flight between signal transmission and reception. The difference is is that the wave frequency are in the Radio frequency area, between 3 MHz and 300 GHz (76 - 81 GHz for automotive RADAR) whereas LiDAR wave frequency is in the infrared area, aproximately 10 000 times higher. This difference in wave frequency translates in sensors property where RADAR :

- Is less senssible to signal attenuation in the athomsphere, allowing higher maximum range and robustness to wheather conditions such as rain or fog.
- Return Doppler velocity of obsctacles.
- Can possibily go through matter depending on the type of RADAR
- Is cheaper than LiDAR, so attractive for industrial applications

However, the RaDAR drawback also include disadvanatges compared to the radar, in fact, it is :

- The accuracy of detections is poor compared to the LiDAR necessiting deeper noise consideration.
- The spatial resolution is weak which makes it complicated to detect small detecttion for weak Radar Cross Section target (RCS) depending on shape, geometry and material.

So if the RADAR appears to be an cheap and robust sensors for hostile envirronement conditions it is complicated to use it whitout signal processing and/or whitout joining other sensors. [1]

An application of the radar benefit : The dynamic tracking

Dynamic objects envirronement need to be identified to

- Remap correctly the map that have changed.
- To avoid the dynamic obstacles.

The dynamic obsctacles are of two types :

- Semi-static where the object does not change its position often (e.g. a door).
- Highly dynamic where the object often move (e.g. a person).

Classically, when range-azimuth sensors are used, dynamic object tracking is performed by comparaison of multiple frame to be able to detect semi-static and high dynamicity. [2]

$$y_r \cos(\theta_i) - x_r \sin(\theta_i) = y_{ICR} \cos(\theta_i) - x_{ICR} \sin(\theta_i) - v_{d_i} \cdot \omega^{-1} \quad (1)$$

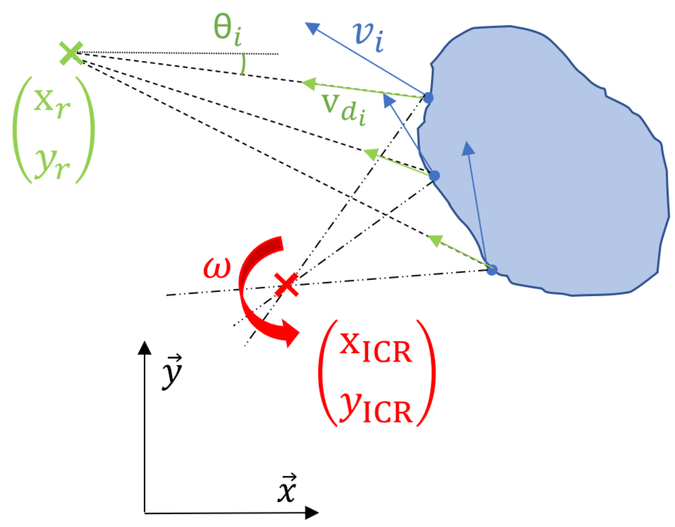

Figure 2. ICR and angular velocity retrieval with distribution of Doppler radial velocity.

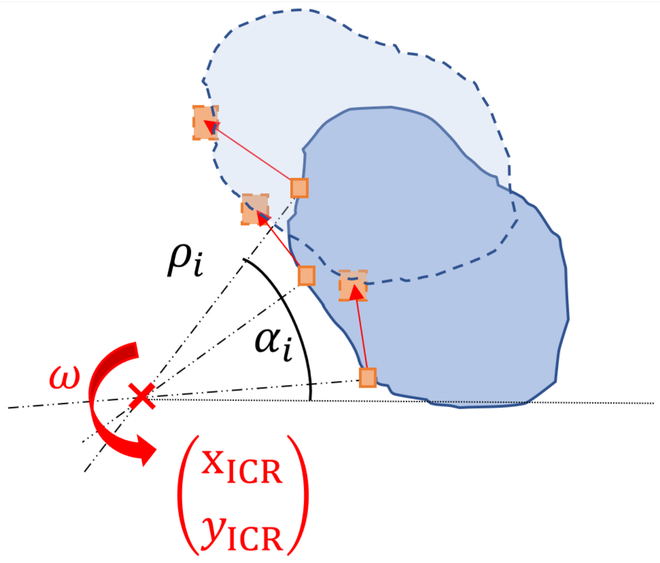

Figure 3. Current measured point projection with ICR and angular velocity estimation.

Research tracks

- To use the Doppler Velocity of the Radar to map the turn of the object.
- To use the shape estimation of the object to obtain conclusion on static map and dynamic obstacles uncertainities.

Bibliography

[1] Kyle HARLOW et al. "A New Wave in Robotics : Survey on Recent MmWave Radar Applications in Robotics". In : *IEEE TRANSACTIONS ON ROBOTICS* 40 (2024), p. 4544-4560.

[2] Dominik KELLNER, Michael BARJENBRUCH et Jens KLAPPSTEIN. "Tracking of extended objects with high-resolution Doppler radar". In : *IEEE Transactions on Intelligent Transportation Systems* 17 (2015), p. 1341-1353.